The CARRE mesh generator

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Outline

1. Overall code procedure
2. Input files
3. How to run CARRE
4. The double-null case

Overall code procedure

The mesh is built as follows. `CARRE`:

1. reads the magnetic field information from `rzpsi.dat`
2. looks for points where $\frac{\partial\Psi}{\partial r} = \frac{\partial\Psi}{\partial z} = 0$
3. asks user to identify from those X- and O-points
4. traces separatrices from X-points to plates
5. follows plates away from strike-point until the field line connecting the plates finds another wall or a corner is turned to find SOL extent
6. asks user for meshing parameters
   - number of cells, first and last interval

   $\Rightarrow$ meshing spacing is quadratic
7. computes minimum and maximum intervals
8. check with user if parameters are OK
9. produces the grid and saves it in `carre.out`

Input files

rzpsi.dat Contains magnetic field information on a (r, z) grid

carre.dat Contains saved parameters from previous runs (optional)

structure.dat Contains wall and target plate information

Magnetic input file : rzpsi.dat

Mesh size is **nr** $\times$ **nz**:

```
$r
 nr= 129
    .11000E+01   .11125E+01   .11250E+01   .11375E+01   .11500E+01
    .11625E+01   .11750E+01   .11875E+01   .12000E+01   .12125E+01
 ...
    .26000E+01   .26125E+01   .26250E+01   .26375E+01   .26500E+01
    .26625E+01   .26750E+01   .26875E+01   .27000E+01
 $z
 nz= 129
  -.12000E+01 -.11812E+01 -.11625E+01 -.11438E+01 -.11250E+01
  -.11063E+01 -.10875E+01 -.10687E+01 -.10500E+01 -.10313E+01
...
    .10500E+01   .10687E+01   .10875E+01   .11063E+01   .11250E+01
    .11438E+01   .11625E+01   .11813E+01   .12000E+01
 $psi
    .35440E+01   .35342E+01   .35245E+01   .35148E+01   .35050E+01
    .34952E+01   .34852E+01   .34750E+01   .34647E+01   .34540E+01
...
  -.14777E-01 -.72110E-01 -.12969E+00 -.18751E+00 -.24552E+00
  -.30373E+00 -.36209E+00 -.42058E+00 -.47918E+00 -.53785E+00
  -.59658E+00 -.65532E+00 -.71405E+00 -.77274E+00 -.83136E+00
  -.88986E+00
```

structure.dat

```
$structures
stretched upper right divertor plane
  5
1.4000          1.1900
2.0000          1.1900
2.0000          1.2000
1.4000          1.2000
1.4000          1.1900
stretched upper left divertor plane
  5
1.2000           0.6000
1.2500           0.6000
1.2500           1.2000
1.2000           1.2000
1.2000           0.6000
stretched lower right divertor plate
  5
1.4000           -1.2000
2.0000           -1.2000
2.0000           -1.1900
1.4000           -1.1900
1.4000           -1.2000
stretched lower left divertor plate
  5
1.2000           -0.6000
1.2500           -0.6000
1.2500           -1.2000
1.2000           -1.2000
1.2000           -0.6000
$end
```

1. Name of a structure
2. Number of points
3. List of corner coordinates
4. Last point must be same as first for closed structures
5. Target plates and walls must be **separate** structures

Within the automated interface from Kukushkin

```
cd $SOLPSTOP/src/Carre
carre -
```

And accept all default choices.

When that fails: by hand!

```
cd $SOLPSTOP/src/Carre
$OBJECTCODE/carre
```

Carre will show you something like:

```
      List of points where grad-psi vanishes:
    -------------------------------------------
  #       R=             Z=
  1       1.55818E+00   -8.50603E-01
  2       2.01654E+00    0.00000E+00
  3       2.01654E+00    0.00000E+00
  4       1.55818E+00    8.50603E-01
   0   to stop selection
  -1   for a limiter configuration
  For a divertor configuration, select X point(s)
```

Then type the X-point indices (here 1 and 4). If there is only one X-point, type the X-point index and then 0. It then asks you for the O-point index (here 2 or 3).

Second stage : grid parameters

You then get output of the form:

```
        SEPARATRICES

 #   nptseg(i)   lg(i)      deltp1(i)   deltpn(i)     dpmin       dpmax
===========================================================================
 1      11      4.1672E-01  1.0648E-02  7.2815E-02  1.0648E-02  7.2815E-02
 2       9      3.9781E-01  2.1167E-02  8.4827E-02  2.1167E-02  8.4827E-02
 3      21      5.0402E+00  1.5575E-01  1.5575E-01  1.5575E-01  3.0774E-01


        RADIAL DISTRIBUTIONS
            Distribution in psi: repart=    2

region  npr(i)    width     deltr1(i)   deltrn(i)     drmin       drmax
===========================================================================
 1       7      9.4934E-03  4.8897E-04  2.7592E-03  4.8897E-04  2.7592E-03
 2       7     -2.9324E-02 -2.1975E-03 -1.2400E-02 -1.2400E-02 -1.3444E-03


        CENTRAL REGION: region i=  3     pntrat max.=1.18996003

pntrat  npr(i)    width     deltr1(i)   deltrn(i)     drmin       drmax
===========================================================================
 .3138    7    -4.2922E-02 -1.4986E-03 -8.4565E-03 -9.6763E-03 -1.4986E-03


        GUARD LENGTH FOR EACH DIVERTOR PLATE

tgarde(1)      tgarde(2)
===========================================================================
 .00000          .00000

        RELAXATION PARAMETERS USED TO CONSTRUCT THE MESH

===========================================================================
nrelax        relax          pasmin         rlcept
   50          .200           1.000E-02      1.000E-06

Do you accept (y/n)?
```

If you wish to change something, answer 'n'. The code then prompts you to type the variable name you want to change and its new value.

What do all the variables mean?

The separatrices segments are numbered as follows:

1. From the X-point to the outer (upper) plate
2. From the X-point to the inner (lower) plate
3. From the X-point to itself around the core

The related variables are:

nptseg number of points along separatrix segment

lg length of separatrix segment (in meters)

deltp1 size of first poloidal step along separatrix segment, leaving from X-point

deltpn size of last poloidal step along separatrix segment

dpmin size of minimum poloidal step along separatrix segment

dpmax size of maximum poloidal step along separatrix segment

What do all the variables mean?

The plasma regions are numbered as follows:

1. SOL (Scrape-Off Layer)
2. PFR (private flux region)
3. Core region

The related variables are:

repart = 1 is using a length unit, = 2 if using a psi unit below (strongly recommended)

npr number of radial points (separatrix = 1)

deltr1 size of first radial step away from the separatrix

deltrn size of last radial step in region

drmin size of minimum radial step in region

drmax size of maximum radial step in region

pntrat depth of the core region

The target plates are numbered as follows:

1. Outer (upper) plate
2. Left (lower) plate

The related variables are:

tgarde Distance (in meters) over which the mesh adapts to the target plate

What should you watch for

A certain number of sanity checks are in order:

- Are `lg(i)`, `dpmin` and `dpmax` of the same sign for any given segment?
- Are `width`, `drmin` and `drmax` of the same sign for any given region?
- Is `tgarde` smaller than the length of the appropriate separatrix segment?
- Is the mesh refining toward the plates?
- Are step sizes around the X-point about symmetric?
- For B2 grids:
  - Do the PFR and the Core have the same `npr`?

Adjust parameters `nptseg`, `detp1`, `deltpn`, `npr`, `deltr1`, `deltrn`, `pntrat` and `tgarde` as necessary.

The connected double-null case

The output on your screen now looks like:

```
         SEPARATRICES

 #   nptseg(i)   lg(i)      deltp1(i)   deltpn(i)     dpmin       dpmax
===========================================================================
 1       11      3.6789E-01  2.0000E-02  1.0000E-03  1.0000E-03  5.5371E-02
 2       11      1.4427E-01  2.0000E-02  1.0000E-03  1.0000E-03  2.0507E-02
 3       23      2.8968E+00  2.0000E-02  2.0000E-02  2.0000E-02  1.9549E-01
 4       23      2.0167E+00  2.0000E-02  2.0000E-02  2.0000E-02  1.3262E-01
 5       11      3.6789E-01  2.0000E-02  1.0000E-03  1.0000E-03  5.5371E-02
 6       11      1.4427E-01  2.0000E-02  1.0000E-03  1.0000E-03  2.0507E-02

         RADIAL DISTRIBUTIONS
             Distribution in psi: repart=    2

region  npr(i)    width     deltr1(i)   deltrn(i)     drmin       drmax
===========================================================================
 1       18      1.1984E-01  3.0000E-04  6.0000E-04  3.0000E-04  1.1009E-02
 2        9     -9.2222E-02 -5.0000E-04 -1.2500E-03 -1.9190E-02 -5.0000E-04
 3       18      9.7354E-02  7.5000E-04  2.5000E-03  7.5000E-04  8.1945E-03
 4        9     -8.4067E-02 -5.0000E-04 -1.2500E-03 -1.7443E-02 -5.0000E-04

         CENTRAL REGION: region i=  5     pntrat max.=1.11346631

pntrat  npr(i)    width     deltr1(i)   deltrn(i)     drmin       drmax
===========================================================================
 .0750    9     -2.1024E-02 -7.5000E-04 -1.7500E-03 -3.6837E-03 -7.5000E-04

         GUARD LENGTH FOR EACH DIVERTOR PLATE

tgarde(1)      tgarde(2)      tgarde(3)      tgarde(4)
===========================================================================
 .05000          .05000         .05000         .05000

 Number of X points:  2
    x                    y                   psi
    1.40002308E+00       1.00013979E+00      -2.30500581E-04
    1.40002306E+00      -1.00013980E+00      -2.30498287E-04

 Point O:
    1.88943869E+00       0.00000000E+00      -2.92493337E+00
 X points are connected.
```

What changes for a double null?

Numbering for separatrix segments:

1. From inner (upper) X-point to outer (upper) plate
2. From inner (upper) X-point to inner (lower) plate
3. From inner (upper) X-point to outer (lower) X-point, on the outside
4. From inner (upper) X-point to outer (lower) X-point, on the inside
5. From outer (lower) X-point to outer (lower) plate
6. From outer (lower) X-point to inner (upper) plate

Numbering for regions (Connected DN):

1. Outer SOL
2. Upper PFR
3. Inner SOL
4. Lower PFR
5. Core

For disconnected DN, region 1 is the region between the separatrices, and the rest of the numbering proceeds as above.
Numbering for target plates:

1. Upper outer (upper) plate
2. Upper inner (lower) plate
3. Lower outer (lower) plate
4. Lower inner (upper) plate

Additional sanity checks for a double-null B2 grid:

- Are both SOLs of the same depth?
- Do all plates have the same number of radial points across them?